

ENERGY DATA HACKDAYS 2026

Energy Fingerprints

Make the invisible energy system visible.

From 15-minute smart-meter data to explainable household asset intelligence.

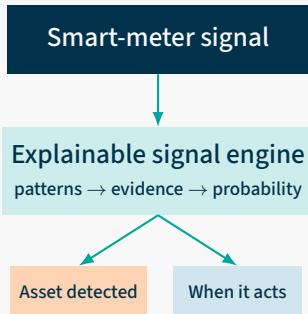
newspaper

AEW Energie challenge

The grid sees one number. We reveal the causes.

A household's net load hides the assets that shape flexibility, infrastructure demand, and solar potential.

- 4 years of anonymised, 15-minute readings
- ~90k households; ~1k labelled customers
- Privacy-preserving: no appliance sensors, no customer intervention



One interpretable workflow, five asset lenses



EV

Sharp switch-on steps in grid import.

Battery

Export displaced past solar noon.

Heat pump

Cold-weather and persistent heating signatures.

Other devices

Repeated distinctive load motifs.

Rooftop PV

Sunny-day midday import depression.

TBD: demo evidence view

EV classifier: a frozen LightGBM benchmark

E22 LIGHTGBM • HELD-OUT TEST

The selected model detects EV-labelled properties from the total net-load profile; it does **not** localise individual charging events.

- 45 features: annual load aggregates, 24-hour mean profile, and windowed multi-meter context.
- Selected by validation PR-AUC among 9 candidates; threshold frozen at 0.392 before the test was opened.
- Test (67 properties; 15 EV-labelled): ROC-AUC 0.67, recall 67%, precision 33%.
- Caveat: “non-EV” means subsidy-register-unlabelled, not confirmed EV-absent.

Held-out evaluation

Actual cohort	Flagged	Not flagged
EV-labelled (GIGI)	10	5
Unlabelled properties	20	32

$$p \geq 0.392; \text{test } n = 67.$$

RULE BASELINE

Flag EV if grid import ever steps up by ≥ 9 kW within one quarter-hour.

ROC-AUC 0.686 n = 342

Actual cohort	Flagged	Not flagged
EV-labelled (GIGI)	49	23
Unlabelled properties	75	195

EV detection has a 2.3 kW blind spot

THE GAP IN THE SPEC

- L1 is 1.4–1.8 kW. L2 starts at 3.6 kW. A Swiss wall socket charges at **2.3 kW** — in between.
- Charging without a wallbox is invisible to every published rule.
- We built the detector for that gap.

RULE BASELINE

At 15 minutes, a boiler and a granny-charger are the *same signal*.

For AEW: needs sub-minute data, or the boiler's tariff-relay channel

AUC 0.49

it finds hot-water boilers

ramp rule: 0.686 $n = 342$

Heat-pump boiler: a physical detector, not a classifier

WPB DETECTOR • LABEL-FREE

The detector identifies heat-pump water-heater candidates from persistent plateaus in 15-minute grid-import data.

- p10 daily envelope: an event must recur on at least 90% of days, rejecting irregular loads such as part-time EV charging.
- Plateau band: 0.25–1.2 kW for 1.5–9 h; it must recur in both January and July.
- Population sweep: 838 of 75,094 households flagged (1.1%) as candidates.
- 4 labelled positives and 0 confirmed negatives: no confusion matrix or accuracy estimate.

Counterfactual recovery test

79.6%

Recovery at 0.3
kW injected

94.2%

Recovery at 0.5
kW injected

Synthetic events test sensitivity only;
they do not establish precision.

Rooftop PV: SolarPrint reads the solar imprint

HISTGRADIENTBOOSTING • 5-FOLD OOF

SolarPrint estimates rooftop-PV probability from grid-import shape; feed-in is used offline for silver labels, never as a feature.

- 23 features: summer daytime near-zero import, evening-to-midday contrast, solar coupling, seasonality, and night base load.
- Positive-Unlabelled learning with Elkan-Noto correction, calibrated to the 12% canton base rate.
- GIGI result: ROC-AUC 0.92; recall 92% at $p \geq 0.5$; 6.6% of the population flagged.
- No confirmed non-PV households: the ranking is stronger than absolute probabilities.

Evaluation snapshot

Actual cohort	Flagged	Not flagged
GIGI PV		
(confirmed)	260	24
Strictly		
unlabelled	4.6k	65.6k

$p \geq 0.5$; silver positives excluded.

RULE BASELINE

Flag PV if the meter ever exported. You cannot inject what you do not generate.

F1 0.920 n = 342

Actual cohort	Flagged	Not flagged
GIGI PV		
(confirmed)	259	22
Unlabelled		
properties	23	38

Battery: a rule baseline, model to follow

PV-CONDITIONED PU DESIGN • NOT YET TRAINED

The intended contrast is battery versus PV-without-battery—not the full population, which would mostly relearn “has PV.”

- Features: flat import, feed-in shaping, ramp smoothing, evening recovery, and energy balance.
- 82 GIGI no-battery households are a held-out audit set, never training negatives.
- The label-blind universe gate needs 95% of known batteries admissible. Measured: **93.7%** (208/222).
- We **halted the pipeline** rather than weaken the gate to produce a number we could not defend.

RULE BASELINE

Of the solar exported in the shoulder months, what share leaves the house *after* solar noon? Flag a battery above 0.50.

ROC-AUC 0.855 n = 242

Actual cohort	Flagged	Not flagged
GIGI battery (confirmed)	170	20
Unlabelled PV homes	21	31

Scored within PV households.

Other-device classifier: turn unknown load shapes into leads

OTHER DISTINCTIVE DEVICE

The model flags repeated, unusual load motifs for review, then groups them into candidate device classes as labels mature.

- Input: recurring event shape, power, duration, timing, and seasonality
- Output: device-class probability or “novel pattern” candidate
- Evidence: a small set of representative events a human can inspect

TBD: cluster of recurring appliance signatures

Evaluation placeholder

Actual cohort	Flagged	Not flagged
Asset present		
(TBD)	TBD	TBD
Asset absent		
(TBD)	TBD	TBD

Define the class and held-out split before reporting results.

Trust is a feature, not a footnote

Explainable

Every score
links to its
evidence.

“Why was this home
flagged?” has a visible
answer.

Honest

Validation is
held out and
limitations stay
visible.

We do not invent accuracy
where labels are missing.

Scalable

One workflow
can score the
full population.

Designed for multi-year,
15-minute data.

TBD: insert one anonymised household evidence card here

From meter data to better grid decisions.

Asset intelligence, activity windows, and evidence—at population scale.

Forecasting

Better demand expectations

Planning

Target infrastructure investment

Flexibility

Find the right homes for the right programme

TBD: closing ask / next experiment

Thank you